

Cumulative Incidence Curves

1.1 Setup

Consider time to event data $\{(T_i, \delta_i)\}$, where time T_i is absolutely continuous and the status δ_i is coded as:

$$\delta_i = \begin{cases} 0 & \text{censoring,} \\ 1 & \text{event,} \\ 2 & \text{competing risk.} \end{cases}$$

Definition 1.1. The **cumulative incidence curve** (CIC) for type-1 events $F_1(t)$ is the probability of experiencing the event before the competing risk by time t :

$$F_1(t) = \mathbb{P}(T \leq t, \delta = 1) = \int_0^t S(u) \lambda_1(u) du,$$

where S is the overall survival function:

$$S(t) = \mathbb{P}(T > t) = e^{-\Lambda_1(t) - \Lambda_2(t)}.$$

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Discussion 1.1. The overall survival function $S(t)$ may be estimated from data of the form $\{(T_i, \delta_i^*)\}$, where:

$$\delta_i^* = \begin{cases} 0 & \text{censoring,} \\ 1 & \text{event or competing risk.} \end{cases}$$

Let $\hat{S}$ denote the Kaplan-Meier estimate of S . The Nelson-Aalen estimate of $\Lambda_1(u)$ is:

$$\hat{\Lambda}_1(t) = \int_0^t \frac{dN_1(u)}{Y(u)},$$

where $Y(u) = \sum_{i=1}^n \mathbb{I}(T_i \geq u)$ is the number at risk and $N_1(u)$ is the counting process for type-1 events:

$$N_1(t) = \sum_{i=1}^n \mathbb{I}(T_i \leq t, \delta_i = 1).$$

The standard estimator of F_1 is:

$$\hat{F}_1(t) = \int_0^t \hat{S}(u) d\hat{\Lambda}_1(u),$$

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1.2 Background

We make use of the following results proved elsewhere.

Proposition 1.1 (Kaplan-Meier to Nelson-Aalen). Let $S(t)$ denote the overall survival function and $\Lambda(t) = \Lambda_1(t) + \Lambda_2(t)$ the overall cumulative hazard. Then:

$$\frac{\sqrt{n}\{\hat{S}(t) - S(t)\}}{-S(t)} = \sqrt{n}\{\hat{\Lambda}(t) - \Lambda(t)\} + o_p(1), \quad (1.1)$$

where $\hat{S}$ is the KM estimator, and $\hat{\Lambda}$ the NA. ◆

Proposition 1.2 (Martingale Expansion of Nelson-Aalen). The NA estimator is expressible as:

$$\sqrt{n}\{\hat{\Lambda}(t) - \Lambda(t)\} = \int_0^t \frac{\sqrt{n}}{Y(u)} dM(u) + o_p(1),$$

where $M(u) = N(u) - \int_0^u Y(s) d\Lambda(s)$ is the counting process martingale.

Likewise, for the cause-specific cumulative hazard:

$$\sqrt{n}\{\hat{\Lambda}_1(t) - \Lambda_1(t)\} = \int_0^t \frac{\sqrt{n}}{Y(u)} dM_1(u) + o_p(1),$$

where $M_1(u) = N_1(u) - \int_0^u Y(s) d\Lambda_1(s)$ is the counting process martingale. ◆

Proposition 1.3. The predictable covariation is a bilinear form:

$$\langle M_1 + M_2, M_1 + M_2 \rangle = \langle M_1 \rangle + \langle M_2 \rangle + 2\langle M_1, M_2 \rangle.$$

The analogous result holds for the optional covariation. ◆

Proposition 1.4. If H_1, H_2 are predictable processes and M_1, M_2 are mean-zero martingales, then the predictable covariation of the corresponding stochastic integrals is:

$$\left\langle \int H_1(t) dM_1(t), \int H_2(t) dM_2(t) \right\rangle = \int H_1(t) H_2(t) d\langle M_1(t), M_2(t) \rangle.$$

The analogous result holds for the optional covariation. ◆

1.3 Martingale Representation

Proposition 1.5.

$$\begin{aligned} \sqrt{n}\{\hat{F}_1(t) - F_1(t)\} &= \int_0^t \sqrt{n}\{\hat{S}(u) - S(u)\} d\hat{\Lambda}_1(u) \\ &\quad + \int_0^t \sqrt{n}S(u)\{d\hat{\Lambda}_1(u) - d\Lambda_1(u)\}. \end{aligned}$$

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Proof.

$$\sqrt{n}\{\hat{F}_1(t) - F_1(t)\} = \int_0^t \sqrt{n}\hat{S}(u)d\hat{\Lambda}_1(u) - \int_0^t \sqrt{n}S(u)d\Lambda_1(u).$$

Adding and subtracting $S(u)$ from the first integrand:

$$\begin{aligned} \int_0^t \sqrt{n}\hat{S}(u)d\hat{\Lambda}_1(u) &= \int_0^t \sqrt{n}\{\hat{S}(u) - S(u) + S(u)\}d\hat{\Lambda}_1(u) \\ &= \int_0^t \sqrt{n}\{\hat{S}(u) - S(u)\}d\hat{\Lambda}_1(u) + \int_0^t \sqrt{n}S(u)d\hat{\Lambda}_1(u). \end{aligned}$$

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Proposition 1.6.

$$\begin{aligned} \int_0^t \sqrt{n}\{\hat{S}(u) - S(u)\}d\hat{\Lambda}_1(u) &= - \int_0^t \sqrt{n}\{\hat{\Lambda}(u) - \Lambda(u)\} \cdot dF_1(u) + o_p(1) \\ &= -\sqrt{n}\{\hat{\Lambda}(t) - \Lambda(t)\}F_1(t) + \int_0^t \sqrt{n}F_1(u)d\{\hat{\Lambda}(u) - \Lambda(u)\} + o_p(1). \end{aligned}$$

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Proof. Since $\hat{\Lambda}_1(\cdot)$ is consistent for $\Lambda_1(\cdot)$:

$$\int_0^t \sqrt{n}\{\hat{S}(u) - S(u)\}d\hat{\Lambda}_1(u) = \int_0^t \sqrt{n}\{\hat{S}(u) - S(u)\}d\Lambda_1(u) + o_p(1).$$

Applying (1.1):

$$\begin{aligned} \int_0^t \sqrt{n}\{\hat{S}(u) - S(u)\}d\Lambda_1(u) &= - \int_0^t \frac{\sqrt{n}\{\hat{S}(u) - S(u)\}}{-S(u)} \cdot S(u)d\Lambda_1(u) \\ &= - \int_0^t \sqrt{n}\{\hat{\Lambda}(u) - \Lambda(u)\} \cdot dF_1(u) + o_p(1). \end{aligned}$$

Integrating by parts:

$$\begin{aligned} \int_0^t \sqrt{n}\{\hat{\Lambda}(u) - \Lambda(u)\} \cdot dF_1(u) &= \sqrt{n} \left[\{\hat{\Lambda}(u) - \Lambda(u)\}F_1(u) \right]_{u=0}^{u=t} - \int_0^t \sqrt{n}F_1(u)d\{\hat{\Lambda}(u) - \Lambda(u)\} \\ &= \sqrt{n}\{\hat{\Lambda}(t) - \Lambda(t)\}F_1(t) - \int_0^t \sqrt{n}F_1(u)d\{\hat{\Lambda}(u) - \Lambda(u)\}. \end{aligned}$$

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Proposition 1.7 (Martingale Representation).

$$\begin{aligned}\sqrt{n}\{\hat{F}_1(t) - F_1(t)\} &= -F_1(t) \int_0^t \frac{\sqrt{n} \cdot dM(u)}{Y(u)} + \int_0^t \frac{\sqrt{n} \cdot F_1(u)}{Y(u)} dM(u) \\ &\quad + \int_0^t \frac{\sqrt{n} \cdot S(u)}{Y(u)} dM_1(u) + o_p(1).\end{aligned}$$

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Proof. From the previous propositions:

$$\begin{aligned}\sqrt{n}\{\hat{F}_1(t) - F_1(t)\} &= -\sqrt{n}\{\hat{\Lambda}(t) - \Lambda(t)\}F_1(t) + \int_0^t \sqrt{n}F_1(u)d\{\hat{\Lambda}(u) - \Lambda(u)\} \\ &\quad + \int_0^t \sqrt{n}S(u)\{d\hat{\Lambda}_1(u) - d\Lambda_1(u)\} + o_p(1).\end{aligned}$$

The conclusion follows from applying the martingale expansion of the Nelson-Aalen estimator to $\sqrt{n}\{\hat{\Lambda}(t) - \Lambda(t)\}$, $\sqrt{n}d\{\hat{\Lambda}(t) - \Lambda(t)\}$, and $\sqrt{n}d\{\hat{\Lambda}_1(t) - \Lambda_1(t)\}$.

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Corollary 1.1.

$$\begin{aligned}\sqrt{n}\{\hat{F}_1(t) - F_1(t)\} &= -F_1(t) \int_0^t \frac{\sqrt{n}}{Y(u)} dM_1(u) + \int_0^t \frac{\sqrt{n} \cdot \{1 - F_2(u)\}}{Y(u)} dM_1(u) \\ &\quad - F_1(t) \int_0^t \frac{\sqrt{n}}{Y(u)} dM_2(u) + \int_0^t \frac{\sqrt{n} \cdot F_1(u)}{Y(u)} dM_2(u) + o_p(1).\end{aligned}$$

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Proof. Substituting $M(u) = M_1(u) + M_2(u)$ and $S(u) = 1 - F_1(u) - F_2(u)$ gives the result.

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1.4 Predictable and Optional Variations**Proposition 1.8.** Suppose $n^{-1}Y(t) \xrightarrow{p} y(t)$, then as $n \rightarrow \infty$:

$$\begin{aligned}\langle \sqrt{n}\{\hat{F}_1(t) - F_1(t)\} \rangle &\xrightarrow{p} F_1^2(t) \int_0^t \frac{\lambda_1(u)}{y(u)} du + \int_0^t \frac{\{1 - F_2(u)\}^2 \lambda_1(u)}{y(u)} du \\ &\quad + F_1^2(t) \int_0^t \frac{\lambda_2(u)}{y(u)} du + \int_0^t \frac{F_1^2(u) \lambda_2(u)}{y(u)} du \\ &\quad - 2F_1(t) \int_0^t \frac{\{1 - F_1(u)\} \lambda_1(u)}{y(u)} du \\ &\quad - 2F_1(u) \int_0^t \frac{F_1(u) \lambda_2(u)}{y(u)} du\end{aligned}\tag{1.2}$$

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Proof. Finding the optional variation:

$$\begin{aligned}\langle \sqrt{n}\{\hat{F}_1(t) - F_1(t)\} \rangle &= F_1^2(t) \int_0^t \frac{n}{Y^2(u)} d\langle M_1(u) \rangle + \int_0^t \frac{n\{1 - F_2(u)\}^2}{Y^2(u)} d\langle M_1(u) \rangle \\ &\quad + F_1^2(t) \int_0^t \frac{n}{Y^2(u)} d\langle M_2(u) \rangle + \int_0^t \frac{nF_1^2(u)}{Y^2(u)} d\langle M_2(u) \rangle \\ &\quad - 2F_1(t) \int_0^t \frac{n\{1 - F_1(u)\}}{Y^2(u)} d\langle M_1(u) \rangle \\ &\quad - 2F_1(t) \int_0^t \frac{nF_1(u)}{Y^2(u)} d\langle M_2(u) \rangle + o_p(1).\end{aligned}$$

Substituting $d\langle M_j(u) \rangle = Y(u)\lambda_j(u)du$:

$$\begin{aligned}\langle \sqrt{n}\{\hat{F}_1(t) - F_1(t)\} \rangle &= F_1^2(t) \int_0^t \frac{\lambda_1(u)}{n^{-1}Y(u)} du + \int_0^t \frac{\{1 - F_2(u)\}^2 \lambda_1(u)}{n^{-1}Y(u)} du \\ &\quad + F_1^2(t) \int_0^t \frac{\lambda_2(u)}{n^{-1}Y(u)} du + \int_0^t \frac{F_1^2(u) \lambda_2(u)}{n^{-1}Y(u)} du \\ &\quad - 2F_1(t) \int_0^t \frac{\{1 - F_2(u)\} \lambda_1(u)}{n^{-1}Y(u)} du \\ &\quad - 2F_1(t) \int_0^t \frac{F_1(u) \lambda_2(u)}{n^{-1}Y(u)} du + o_p(1).\end{aligned}$$

Letting $n^{-1}Y(u) \xrightarrow{p} y(u)$ gives the result. ■

Discussion 1.2. By the martingale central limit theorem:

$$\sqrt{n}\{\hat{F}_1(t) - F_1(t)\} \rightsquigarrow W\{\sigma_{\text{CIC}}^2(t)\},$$

where $\sigma_{\text{CIC}}^2(t)$ is the RHS of (1.2). An estimate of the variance is obtained by finding the optional variation:

$$\begin{aligned}\hat{\sigma}_{\text{CIC}}^2(t) &= \left[\sqrt{n}\{\hat{F}_1(t) - F_1(t)\} \right] = F_1^2(t) \int_0^t \frac{dN_1(u)}{n^{-1}Y^2(u)} + \int_0^t \frac{\{1 - F_2(u)\}^2 dN_1(u)}{n^{-1}Y^2(u)} \\ &\quad + F_1^2(t) \int_0^t \frac{dN_2(u)}{n^{-1}Y^2(u)} + \int_0^t \frac{F_1^2(u) dN_2(u)}{n^{-1}Y^2(u)} \\ &\quad - 2F_1(t) \int_0^t \frac{\{1 - F_2(u)\} dN_1(u)}{n^{-1}Y^2(u)} \\ &\quad - 2F_1(t) \int_0^t \frac{F_1(u) dN_2(u)}{n^{-1}Y^2(u)}.\end{aligned}$$

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Area Under the Cumulative Incidence Curve

Example 2.1. Consider the area under the cumulative incidence curve (AUCIC):

$$U(\tau) = \int_0^\tau F_1(t)dt = \int_0^\tau \int_0^t S(u)d\Lambda_1(u).$$

and let $\hat{U}(\tau)$ denote the estimator:

$$\hat{U}(\tau) = \int_0^\tau \hat{F}_1(t)dt = \int_0^\tau \int_0^t \hat{S}(u)d\hat{\Lambda}_1(u).$$

Using the martingale representation of $\sqrt{n}\{\hat{F}_1(t) - F_1(t)\}$, the standardized difference:

$$\sqrt{n}\{\hat{U}(\tau) - U(\tau)\} = \int_0^\tau \sqrt{n}\{\hat{F}_1(t) - F_1(t)\}dt$$

is expressible as:

$$\begin{aligned} \int_0^\tau \sqrt{n}\{\hat{F}_1(t) - F_1(t)\}dt &= - \int_0^\tau F_1(t) \int_0^t \frac{\sqrt{n}}{Y(u)} dM_1(u)dt + \int_0^\tau \int_0^t \frac{\sqrt{n}\{1 - F_2(u)\}}{Y(u)} dM_1(u)dt \\ &\quad - \int_0^\tau F_1(t) \int_0^t \frac{\sqrt{n}}{Y(u)} dM_2(u)dt + \int_0^\tau \int_0^t \frac{\sqrt{n}F_1(u)}{Y(u)} dM_2(u)dt + o_p(1). \end{aligned}$$

The predictable variation is:

$$\begin{aligned} \langle \sqrt{n}\{\hat{U}(\tau) - U(\tau)\} \rangle &= \int_0^\tau F_1^2(t) \int_0^t \frac{\lambda_1(u)}{n^{-1}Y(u)} du dt + \int_0^\tau \int_0^t \frac{\{1 - F_2(u)\}^2 \lambda_1(u)}{n^{-1}Y(u)} du dt \\ &\quad + \int_0^\tau F_1^2(t) \int_0^t \frac{\lambda_2(u)}{n^{-1}Y(u)} du dt + \int_0^\tau \int_0^t \frac{F_1^2(u) \lambda_2(u)}{n^{-1}Y(u)} du dt \\ &\quad - 2 \int_0^\tau F_1(t) \int_0^t \frac{\{1 - F_2(u)\} \lambda_1(u)}{n^{-1}Y(u)} du dt \\ &\quad - 2 \int_0^\tau F_1(t) \int_0^t \frac{F_1(u) \lambda_2(u)}{n^{-1}Y(u)} du dt + o_p(1). \end{aligned}$$

Let $\sigma_{\text{AUCIC}}^2(\tau)$ denote the limit in probability of the predictable variation as $n \rightarrow \infty$. By the martingale central limit theorem:

$$\sqrt{n}\{\hat{U}(\tau) - U(\tau)\} \rightsquigarrow W\{\sigma_{\text{AUCIC}}^2(t)\}.$$

An estimate of the asymptotic variance is obtained from the optional variation:

$$\begin{aligned} \hat{\sigma}_{\text{AUCIC}}^2(\tau) &= \int_0^\tau \hat{F}_1^2(t) \int_0^t \frac{dN_1(u)}{n^{-1}Y^2(u)} dt + \int_0^\tau \int_0^t \frac{\{1 - \hat{F}_2(u)\}^2 dN_1(u)}{n^{-1}Y^2(u)} dt \\ &\quad + \int_0^\tau \hat{F}_1^2(t) \int_0^t \frac{dN_2(u)}{n^{-1}Y^2(u)} dt + \int_0^\tau \int_0^t \frac{\hat{F}_1^2(u) dN_2(u)}{n^{-1}Y^2(u)} dt \\ &\quad - 2 \int_0^\tau \hat{F}_1(t) \int_0^t \frac{\{1 - \hat{F}_2(u)\} dN_1(u)}{n^{-1}Y^2(u)} dt \\ &\quad - 2 \int_0^\tau \hat{F}_1(t) \int_0^t \frac{\hat{F}_1(u) dN_2(u)}{n^{-1}Y^2(u)} dt. \end{aligned}$$

